Geometric Design and Safety Analysis of Roadways

Technical report | Transportation Engineering | Academic project

Review status: Technical verification required before design conclusions are claimed.

1. Executive Summary

Four supplied spreadsheets were organized into a repository covering traffic signal timing, design traffic estimation, axle-load damaging effects, and rigid pavement fatigue calculations. Selected data extracts and illustrative charts are included. Formula-level issues were identified in the fatigue worksheet and need to be resolved before its displayed total fatigue damage is relied upon.

2. Source Workbooks and Scope

Workbook (organized name)	Subject
traffic-signal-timing-four-legged-intersection.xlsx	Approach flows, saturation flow, cycle and green-time allocation
design-traffic-calculations.xlsx	Traffic growth, design traffic and axle repetitions
axle-load-damaging-effect.xlsx	Fourth-power damaging-effect calculations
design-axle-and-fatigue-damage.xlsx	Design axle calculations and rigid pavement fatigue worksheet

The full project title also refers to geometric design and safety analysis, but no clear calculations for superelevation, transition curves, sight distance, or horizontal/vertical alignment were identifiable in the four workbooks supplied.

3. Methods Represented

Design traffic: The workbook uses a cumulative traffic-growth relationship of the form N = 365 × (((1 + r)^n – 1) / r) × A. Confirm the definition of A and the interpretation of r = 0.075 as 7.5% per year.

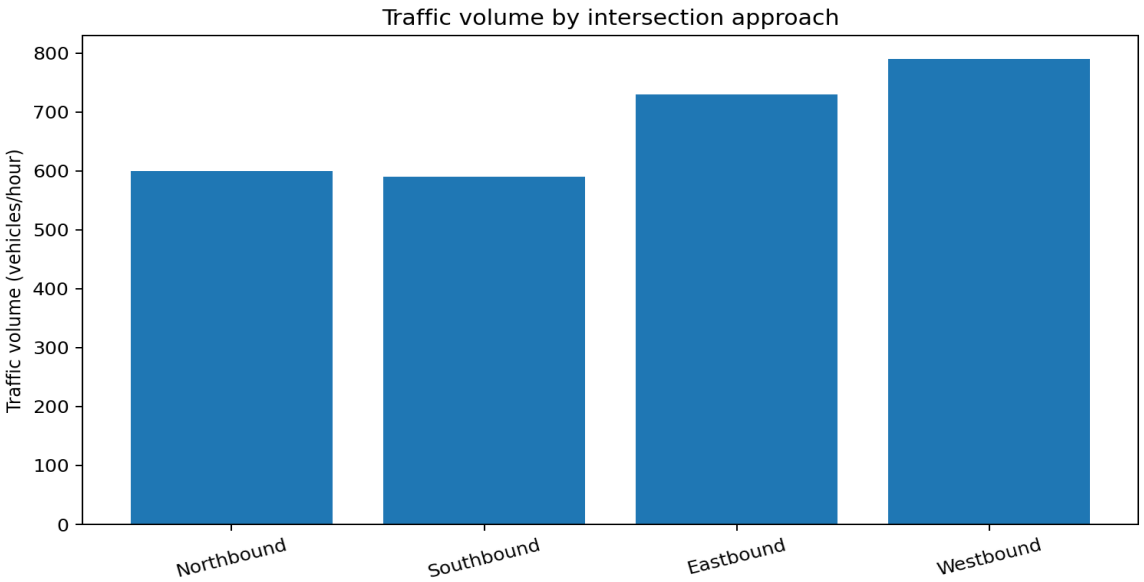
Axle-load damaging effect: The workbook applies a fourth-power relationship D = (W / Wref)^4. The reference load and axle grouping must be confirmed for each formula block.

Signal timing: Approach width and traffic volume feed saturation-flow and volume-to-saturation-flow ratios, followed by cycle and green-time allocation. Lost time, amber and all-red assumptions must be consistent.

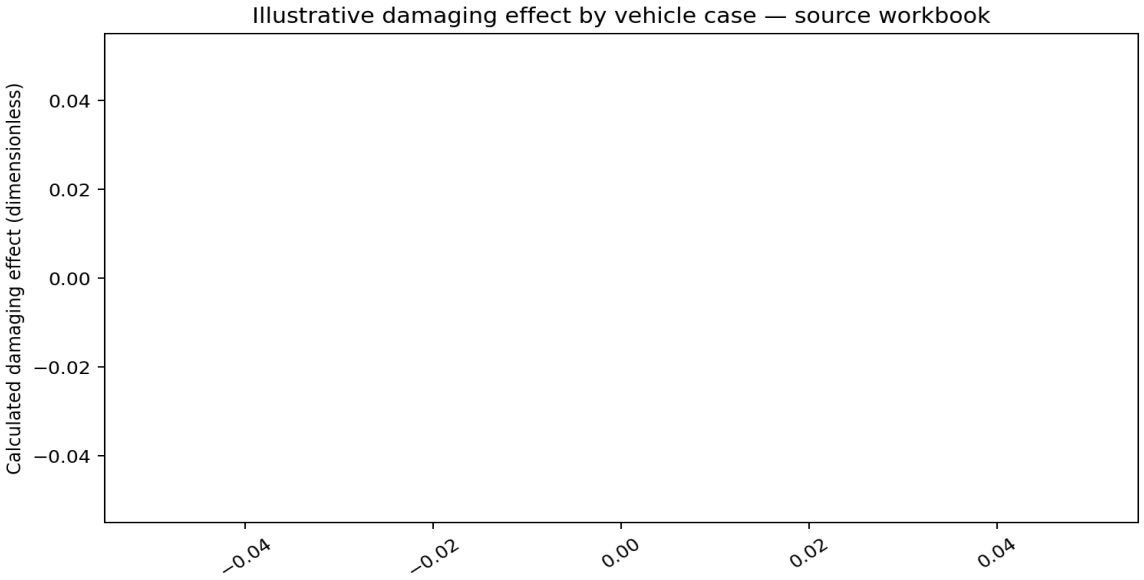
Rigid pavement fatigue: The worksheet groups axle loads into classes and calculates repetitions, flexural stress, stress ratio, allowable repetitions, and fatigue damage. BUC and TDC blocks are both present.

4. Illustrative Charts

Approach traffic volumes recorded in the signal workbook:



Illustrative damaging-effect totals from the first vehicle-case block in the axle-load workbook:



These figures reproduce workbook values and are descriptive only; they are not independent design validation.

5. Technical Quality-Assurance Register

ID	Priority	Finding and required action
QA-01	Critical	Several flexural-stress formulas use shifting condition references (e.g., E19 checks B5, E20 checks B6, E21 checks B7) rather tha
QA-02	Critical	The displayed total fatigue damage in B97 sums the left-hand BUC block only (H65+H80+H95). The right-hand TDC block is prese
QA-03	High	The workbook shows effective green time of 41 s and calculated phase greens around 18 s and 22 s (sum ≈39.6 s). Confirm lost ti
QA-04	High	Growth rate is entered as 0.075 with a label indicating percent. Confirm this is intended to mean 7.5% per year and that the equati
QA-05	High	The workbook uses different denominators (65 kN and 80 kN) across the first data block, and 65 kN for both axle positions in the s
QA-06	Medium	Check whether axle proportions sum to 1.00 and clarify whether the lateral placement factor 0.25 is intended as a multiplier or per
QA-07	Medium	Add units, source references, design assumptions, version/date, and a legend distinguishing inputs from formula outputs. Correct s
QA-08	Scope	The four supplied workbooks do not provide clear calculations for superelevation, transition curves, sight distance, or horizontal/ve

6. Recommended Validation Workflow

• Preserve originals and audit copies separately.

• Identify exact source standards, editions, clauses, and equations.

• Verify units and percentage conventions.

• Audit formula references row-by-row, including absolute and relative references.

• Recalculate using a spreadsheet engine and independently check representative cases.

• Document discrepancies and close all critical QA items before publishing final results.

7. Limitations and Disclaimer

This is an academic learning and portfolio package, not a construction-ready design. Spreadsheet outputs have not been independently certified, exact standards references remain to be confirmed, and the CSV files contain selected cached outputs. Use in real projects requires independent engineering verification and review by a qualified professional.